

Emotionally-Modulated Trading Agent: Implementation Report

1. Introduction

I built the trading agent from my proposal in MeTTa and ran it through PeTTa, along with the baseline agent Nahom suggested for comparison. Everything described here actually runs; all the numbers and figures in this report come straight from the run logs.

The setup is the one from the proposal. The MetaMo agent trades a single asset over a series of prices, choosing between buy, hold, and sell each step. Its appraisal step updates the six OpenPsi modulators from what the market just did, the two overgoals (individuation for protecting capital, transcendence for chasing gains) shape both the appraisal and the decision scoring, and the state updates through the incremental blend so it changes gradually instead of jumping. The emotion labels (fear, confident, sad, neutral) are computed from the modulators each step, the same way as in my earlier emotion task. The baseline is the fixed-rule momentum trader we agreed on: it buys when the price went up last step and sells when it went down, forever, with no internal state at all. Both agents trade under exactly the same market rules, so any difference in what they do comes from the MetaMo machinery.

I tested it three ways. First on a hand-made price series where the market rises, crashes, and then genuinely recovers. Second on a nastier version of the same series where the recovery is a trap: a short bounce followed by a second, deeper crash. And third, because two hand-made series are not really evidence, I ran both agents over thirty randomly generated markets and compared the distribution of their final values.

The short version of the results: on the first series the baseline finished ahead (96.5 against 93.5), because its blind re-buying after the crash happened to get rewarded. On the trap series the MetaMo agent finished ahead (93.5 against 89.5), because it was still cautious from the first crash and refused the fake bounce that the baseline bought into. And over the thirty random markets, the MetaMo agent ended higher on 24 of them, its worst outcome was 93.4 against the baseline's 88.3, and on the strongest bull markets it still kept up with or beat the baseline. So the honest claim is not that the MetaMo agent earns more. It is that its outcomes are much more stable: the baseline's result swings with market luck, while the MetaMo agent's barely does. That is basically the homeostatic stability idea from the papers showing up as actual money.

One moment from the runs is worth pointing out because it captures the whole thing. On the trap series, at the bottom of the first crash, the agent's securing and threshold modulators had climbed high enough that the emotion came out as fear, and it sold its last share. Then during the fake bounce its mood actually recovered (the emotion label went back to confident), but the individuation overgoal was still elevated from the capital loss, so the buy action kept scoring badly and it held. Two steps later the trap sprang and it owned nothing. The baseline bought

the bounce and rode it down. That one sequence is the clearest demonstration I have that the motivational layer and the mood layer are doing separate, useful work.

2. How the Comparison Was Set Up

The point of the whole setup was to make sure that any difference between the two agents could only come from the MetaMo machinery, and not from anything else. So both agents trade under identical conditions: the same price series, the same starting cash (100.0), the same three actions, and the same market rules. Those rules live in one shared piece of code, so there is no possibility of the two agents accidentally playing by different rules. A buy that the agent cannot afford, or a sell with no shares to sell, safely turns into a hold for both of them, and the logs print the action that actually happened rather than what the agent intended.

The baseline is deliberately dumb, and that is its job. It is a momentum rule: buy if the price rose last step, sell if it fell, hold if it was flat. It has no memory of how it has been doing, no internal state, and it never changes. I picked this (from the options Nahom and I discussed) because it is the simplest agent that still actively trades, which makes it a clean control: whatever the MetaMo agent does differently has to come from its modulators and overgoals, because that is literally the only difference between them.

For the markets, I used three levels of testing, each one fixing a weakness of the previous one.

The first is series A, a hand-made twelve-step series where the price rises from 10 to 14, crashes to 9, and then genuinely recovers to 12. This is the friendly case, and honestly it is a case where blind momentum should do well, since the recovery is real and re-buying it pays off. I included it partly for that reason: I did not want to only test on markets that favour my agent.

The second is series B, which is identical to series A through the crash, but the recovery is a trap. The price bounces from 9 up to 11, and then falls again to 7. I designed this one specifically to punish blind re-buying, and I want to be upfront that it is designed that way. On its own it would prove nothing, since I could obviously construct a market where caution wins. Its value is only in contrast with series A: the same two agents, one market where recklessness is rewarded and one where it is punished, and you can watch how each agent's outcome depends on which world it happens to be in.

The third level is the one that actually carries the evidence, because the first two are both hand-made. I generated thirty random markets, each twenty-five steps long, using a small seeded random number generator I wrote in MeTTa (so the runs are reproducible: the same seed always produces the same thirty markets). The price steps are drawn from four moves that sum to zero, so the generated markets have no built-in upward or downward drift and neither agent gets a structural advantage. Both agents were run over all thirty markets and I collected the final values, looking at the worst case, the mean, and the best case for each. Neither agent's parameters were tuned on these markets; the MetaMo agent's constants were set while working

with series A and B, and the batch markets were only run afterward. That matters, because it means the batch is a fair test of whether the behavior generalizes rather than something I fitted.

One implementation detail from the batch is worth admitting because it nearly fooled me. My first version of the random generator had a classic bug: I was taking the low bits of a linear congruential generator, and the low bits of an LCG cycle with a short period, so every "random" market was secretly one of only two shapes. The batch results looked wonderful and were completely meaningless. The fix was one line (routing the seed through an odd prime modulus first), but I am mentioning it because the lesson was bigger than the bug: the batch summary looked plausible both before and after the fix, and I only caught it by inspecting the individual runs and noticing the values repeating. Checking the evidence generator, not just the evidence, is now something I will do by default.

3. Results

3.1 Series A: the friendly market

On the series with a real recovery, the baseline finished at 96.5 and the MetaMo agent at 93.5. Both agents behaved identically through the rise and the crash: they bought on the way up from 10 to 14, and sold on the way down to 9. The entire difference happened after the bottom. The baseline's rule saw the price rising again and immediately re-bought at 10 and 11, riding the recovery back up. The MetaMo agent held instead, and its value stayed flat at 93.5 for the rest of the run.

The reason it held is visible in the logs. At the bottom of the crash (step 9) its individuation overgoal had climbed to about 0.60, and its securing and threshold modulators were high enough that the emotion came out as fear. During the recovery its mood actually improved, the emotion label went back to confident by step 11, but `g_Ind` was still elevated at around 0.59 to 0.60, and in the decision scoring that keeps the buy action penalized. So the agent was in the odd but very human state of feeling better while still not being willing to commit money again. Figure 1 shows the two value curves separating after the bottom, and figure 2 shows the overgoal crossover around step 6 where the agent's overall stance flipped from gain-chasing to capital protection.

One more thing worth pointing at in the series A log: `g_Ind` does not freeze after the crash. It peaks at 0.602 at step 10 and then drifts down, 0.598, then 0.590, as the reference point the agent judges itself against slowly adapts and the rising market cools its caution. On a longer recovery it would eventually re-enter. This mattered because an earlier version of the agent had exactly this bug, the caution locked at one value forever, and section 4 describes the fix.

So on this market, honestly, the baseline won, and it won because recklessness happened to be the right move. That is fine, and it is why series B exists.

3.2 Series B: the trap

On the trap series the finals flip: the MetaMo agent kept 93.5 and the baseline fell to 89.5. Everything up to step 11 plays out the same as series A, including the fake bounce from 9 to 11. The baseline bought the bounce at 10 and again at 11, exactly as its rule requires. The MetaMo agent held through both steps for the same reason as before, its caution was still elevated. Then at step 12 the trap sprang, the price fell to 9 and kept falling to 7, and the baseline sold its way down at a loss while the MetaMo agent, holding nothing but cash, lost nothing at all. Its emotion through the second crash came out as fear again, which is fitting, but the practical point is that it was afraid while sitting safely in cash.

Figure 6 is the picture of this whole story. The top panel shows the price path with both agents' actions on it, and the steps where the two agents disagreed are shaded. There are no shaded steps before the bounce, both agents saw the same market and made the same choices. Every shaded step after it is the divergence, and the bottom panel shows the caution drive g_Ind sitting above neutral through exactly those steps. Figure 4 puts the two series side by side and makes the summary visible in one glance: the MetaMo agent's final value is 93.5 in both worlds, while the baseline's swings from 96.5 to 89.5 depending on which world it happens to be in.

3.3 The batch: thirty random markets

The batch is where the claim stops being a story about two hand-made series. Over thirty random markets that neither agent was tuned on, the results were:

The baseline: worst 88.3, mean about 100.2, best 156.4.

The MetaMo agent: worst 93.4, mean about 102.1, best 157.6.

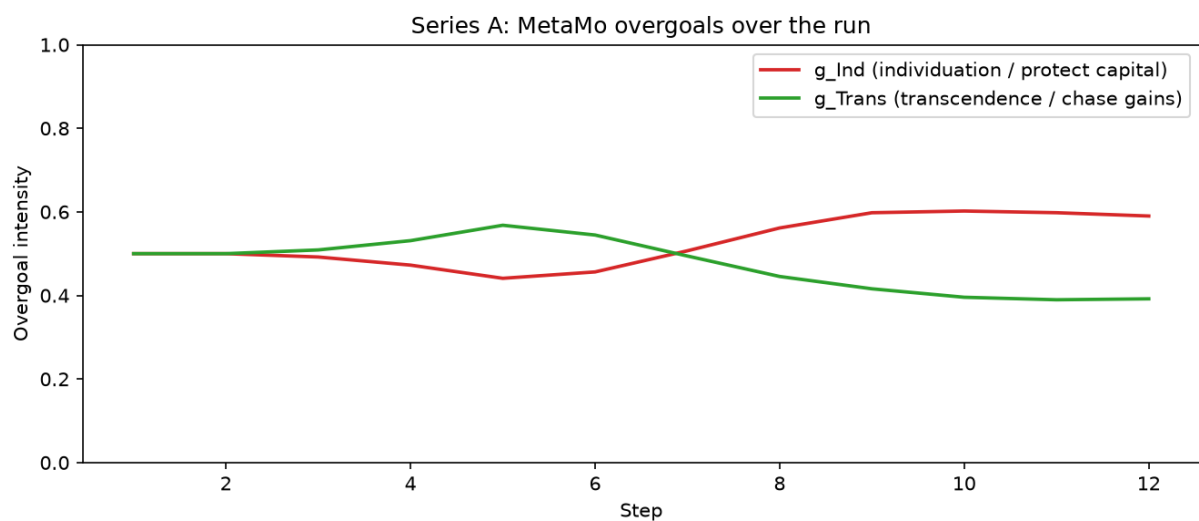
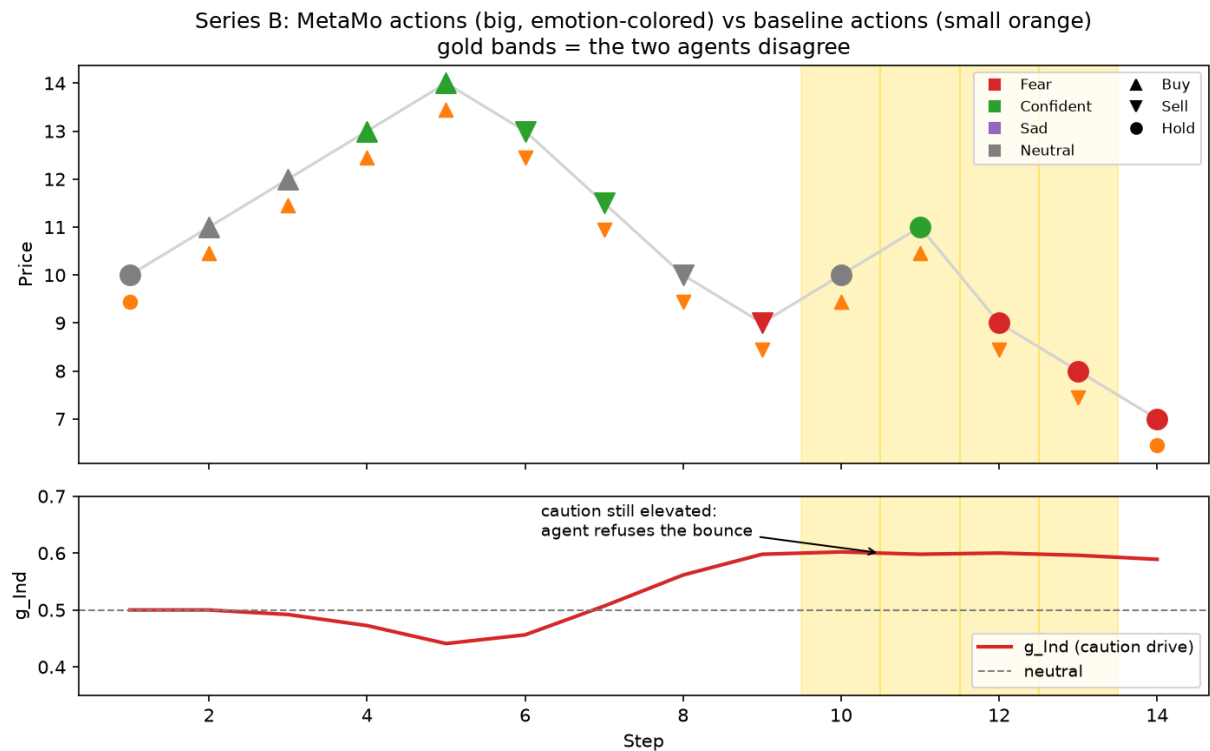
The MetaMo agent finished higher on 24 of the 30 markets. Its worst case was about five points better than the baseline's, which is the number I care most about, since the whole argument is that the emotional machinery works as risk management. And the part that surprised me: it did not pay for that protection with upside. On the two strongest bull markets in the batch it actually finished ahead of the baseline as well, 147.7 against 140.4 on one and 157.6 against 156.4 on the other. I expected the cautious agent to lag in bull markets and it did not, I think because the momentum baseline whipsaws (it sells on every single down-tick, even inside a strong uptrend) while the MetaMo agent's blended, gradual state changes make it steadier in both directions.

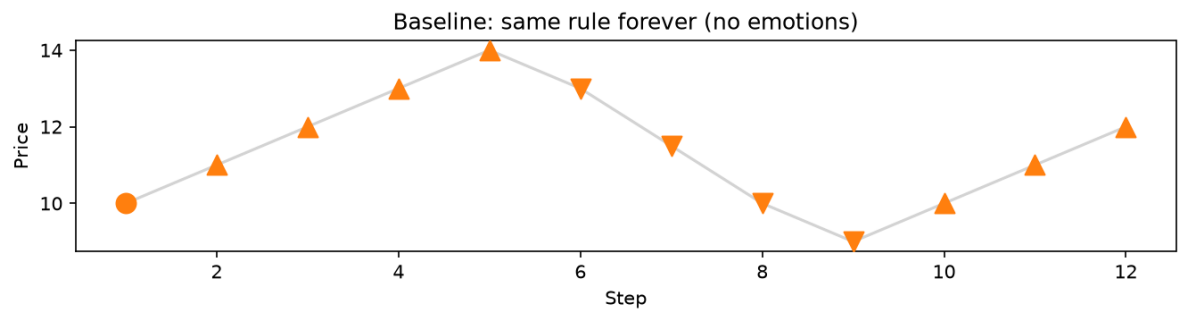
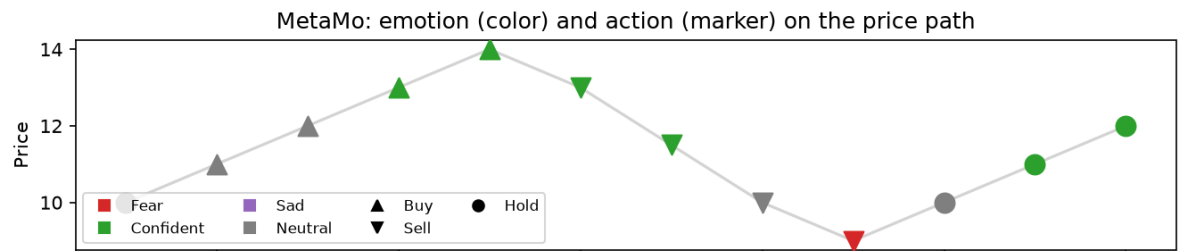
Figure 5 shows all thirty markets as one dot each, baseline final value across, MetaMo final value up. Most dots sit above the equal-outcome diagonal, and the dotted lines mark the two worst cases, with the MetaMo floor clearly above the baseline floor.

3.4 What the results do and do not show

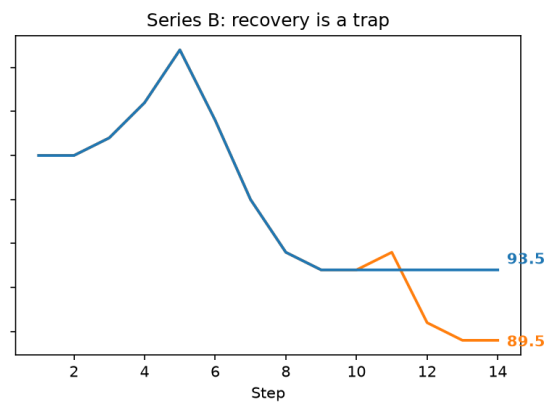
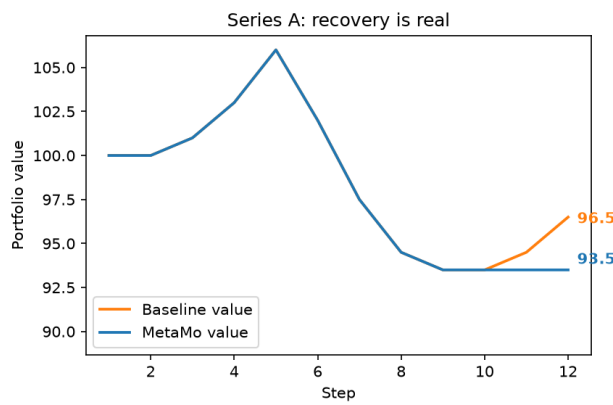
To state the claim precisely: these results do not show that the MetaMo agent earns more money. On individual markets the baseline sometimes beats it, including series A. What they show is that the MetaMo agent's outcomes are far less dependent on market luck: a better worst case, a slightly better mean, matched best cases, and near-identical results across two hand-made worlds designed to reward opposite temperaments. The baseline is gambling and sometimes wins; the MetaMo agent is managing its exposure to its own recent experience, which is exactly what the homeostatic stability principle in the papers says a motivational

system should do. Seeing that principle turn into a five-point difference in worst-case portfolio of the week.value is, to me, the main result

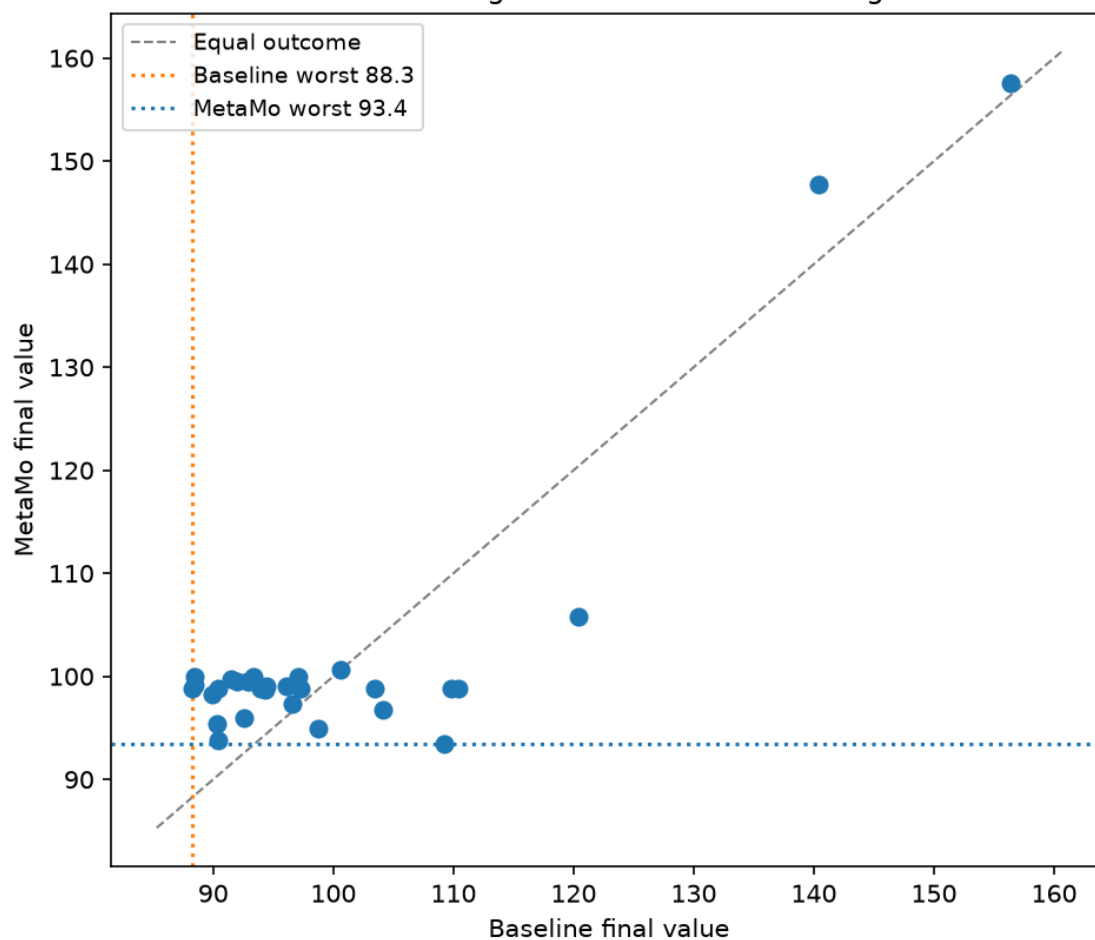




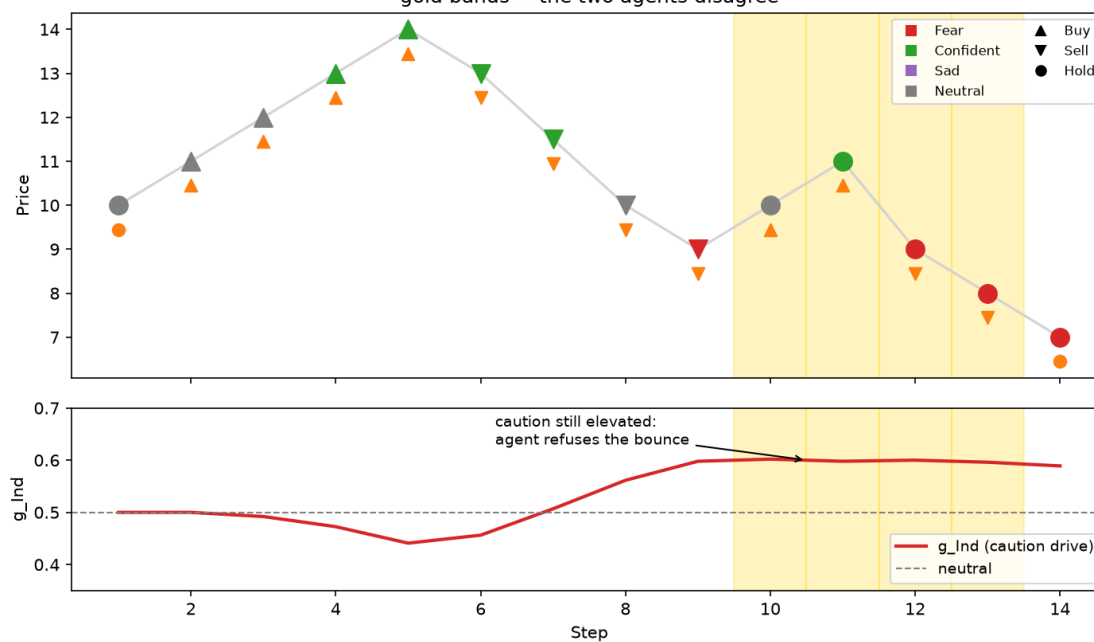
Same agents, two markets: MetaMo's outcome is stable, the baseline's depends on luck



30 random markets: one dot per market
above the diagonal = MetaMo ended higher



Series B: MetaMo actions (big, emotion-colored) vs baseline actions (small orange)
gold bands = the two agents disagree



4. What I Changed Along the Way

The agent that produced the results in section 3 is not identical to the one in my proposal. Three things changed during the week, and I think all three are worth describing, partly because the changes themselves are interesting and partly because two of them came from catching problems in my own first version.

4.1 The overgoal freeze, and the habituation fix

The biggest change was to how the overgoals work. In my first version, `g_Ind` and `g_Trans` were computed fresh each step from a simple formula over the ratio of current portfolio value to starting value. That worked, but it had a problem I only noticed by reading the run logs: once the agent sold everything after the crash, its portfolio value stopped changing, so the ratio stopped changing, so the overgoals locked at one value forever. The log showed `g_Ind` frozen at exactly 0.565 for every remaining step. Behaviorally that meant the agent would still be "cautious from that one crash" a thousand steps later. That is not caution, it is trauma with no way to heal, and it also meant the agent could never re-enter a genuinely recovering market.

The fix has two parts, and both come from making the overgoals real state instead of a formula. First, each step the overgoals track a target driven by capital health, rather than jumping straight to it, and a market move that contradicts a drive cools that drive slightly toward neutral: a rising market cools the caution, a falling market cools the enthusiasm. Second, and this is the part I like, the reference point the agent measures its capital against is no longer fixed at the starting 100. It drifts slowly toward the current value, so after enough time at a new level, that level starts to feel normal. This is basically habituation, the way a person's sense of "how I'm doing" adapts to their situation. With both pieces in place, the logs show `g_Ind` peaking at 0.602 after the crash and then drifting down step by step instead of freezing, and on a long enough recovery the agent regains the will to trade. I also made the target mapping steeper at the same time, because in the first version the overgoals only ever moved within about thirteen percent of their possible range, which meant they were nudges rather than drives.

4.2 Tuning the fear threshold

The second change was small but honest to report. The emotion labels come from thresholds over the modulators, and in my first run the fear label never fired: at the very bottom of the crash, securing and threshold reached about 0.61, just under my 0.65 cutoff, so the most dramatic moment of the run was labelled neutral. I lowered the cutoff to 0.60 and fear now fires exactly at the crash bottom and during the second crash of the trap series. I want to be clear about what this was: the underlying behavior never changed, the agent was already selling defensively at that moment. What I tuned was where the label boundaries sit, so that the names line up with the regimes they are meant to describe. This is the kind of adjustment my proposal's phase 6 predicted ("getting the agent to shift at sensible moments will probably take some adjusting"), and it took one line.

4.3 The random generator bug

The third change I already described in section 2, but it belongs in this list: the first version of my batch generator produced only two distinct market shapes because I was using the low bits of a linear congruential generator, which cycle with a short period. The one-line fix routes the seed through an odd prime modulus before taking the step. I am repeating it here because of the order in which things happened: the broken batch produced a summary that looked excellent, and I nearly reported it. The habit I am taking away is to always look at the individual runs behind any summary statistic I produce.

4.4 A clarification I settled during the week: emotions are readouts

One question that came up while testing deserves its own note, because I expect it is the first question anyone skeptical will ask: does the emotion actually do anything? The precise answer is that the emotion labels do nothing, and that is by design. The causal chain is market event, then modulators and overgoals, then decision: securing and approach sit directly inside the scoring equation, and g_Ind gates the risk penalty. The emotion is then computed from the modulators as a description of the state they are in. So if you deleted the emotion file entirely, the agent would trade identically and only the logs would change. This is not a weakness of my implementation, it is the OpenPsi position implemented faithfully: emotions emerge from the dynamics rather than being hard-coded rules, and fear is the name of the state that causes the retreat, not the cause of it. Wiring the fear label itself into the decisions would have been easy and would have looked more impressive, and it would have been exactly the "if threat then fear then flee" rule structure the papers argue against.

5. Conclusion

This week I took the trading agent from my proposal and actually built it: the MetaMo agent in MeTTa with OpenPsi modulators, the two overgoals, the appraise-decide-blend cycle, and emotions computed on top, alongside the fixed-rule baseline we agreed on for comparison. Everything runs in PeTTa, the market rules are shared between both agents, and all the results come from the logs of real runs.

The main finding is the one I did not fully expect when I wrote the proposal. The emotional machinery does not make the agent trade better in the sense of earning more, and I do not think it should be sold that way. What it does is make the agent's outcomes stable across worlds: the same final value on a market that rewards recklessness and a market that punishes it, a five-point-better worst case over thirty random markets, and no lost upside in the strongest bull runs. The baseline's result is a function of market luck; the MetaMo agent's is a function of its own regulation. That is the homeostatic stability principle from the papers, not as a theorem but as a number in a portfolio.

Along the way I also learned things I would not have gotten from the proposal stage alone: that overgoals need to be state with decay and habituation rather than a formula, or the agent's caution freezes into permanent trauma; that a summary statistic can look perfect while the

generator behind it is broken; and that keeping the emotion labels causally inert, which felt like a weakness when I first noticed it, is actually the faithful implementation of the theory.

If this direction looks right, my suggested next steps in order would be a sensitivity check on the constants using the existing batch infrastructure, a second baseline to strengthen the comparison, position sizing to make the caution continuous, and then the commutation experiment to connect the implementation back to the formal property the first paper uses this exact use case to illustrate.